CERTIFICATE IN STRUCTURAL STEEL DETAILING USING TEKLA - 40 HOURS

INSTITUTE OF INDUSTRIAL DESIGN

S.NO	Session	Topics	Type	Duration (min)	Credits
1	Session 1	**Introduction**	Theory/Demo	120	
		Overview of steel structures			
		Standards and specifications as per IS 800			
		Overview of connections in Industry.			
		User Interface			
		Tekla User Interface			
		Shortcuts			
		Views			
		Grids			
		Grid Editing			
		Types of Grids			
		Practice 1: Create a radial grid with the given specific dimension & Label it.	**Practice**	**120**	**5**
2	Session 2	Work area	Theory/Demo	120	
		Steel Modeling			
		Beam			
		Column			
		Poly Beam			
		Curved Beam			
		Twin Profile			
		Bracings			
		Plate			
		Purlin			
		Contour plate			
		Practice 2: Create a model with steel parts.	**Practice**	**120**	**5**
3	Session 3	**Modifying Parts**	Theory/Demo	60	
		Move Special			
		Linear			
		Rotate			
		Mirror			
		To Another Plane			
		To Another Object			
		Copy Special			
		Linear			
		Rotate			
		Mirror			
		To Another Plane			
		To Another Object			
		Cuts & Fits	Theory/Demo	60	
		Cut Part with Line			
		Cut part with Polygon			
		Cut Part with another part			
		Fit Part End			
		Filters			
		Selection Filter			
		View Filter			
		Practice 3: Apply filters for various steel parts and select the part and modify its properties.	**Practice**	**60**	**6**
		Practice 4: Use Cuts & Fits tool, at the joints for further placements of connections.	**Practice**	**60**	**6**
4	Session 4	**Points**	Theory/Demo	60	
		Points along the extension of two picked points			
		Points on line			
		Points parallel to two picked points.			
		Projected points on line			
		Points at intersection of two lines			
		Points at any position			
		Construction Line			
		Construction Circle			
		Bolts	Theory/Demo	60	
		Manual Creation of Bolt			
		Creating Bolts			
		Modifying Bolts			
		Practice 5: Create bolts and modify its properties as per the given diagram	**Practice**	**60**	**6**
		Practice 6: Create points which is to be divided as per the given diagram	**Practice**	**60**	**6**
		Welds	Theory/Demo	60	
		Manual Creation of Weld			
		Creating Welds			
		Modifying Welds			

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		Types of welds			
		Practice 7: Create welds and modify its properties as per the given diagram	**Practice**	**60**	**6**
5	Session 5	**Connections**	Theory/Demo	60	
		Types of Connections			
		Base specifications of connections			
		Detailed Connection			
		Macro Connection			
		Detailed Connection			
		Base plate Connection with Stiffeners			
		Base plate Connection without Stiffeners			
		Practice 8: Create a base plate connections with & without stiffeners and modify its parameters	**Practice**	**60**	**6**
6	Session 6	**Detailed Connection**	Theory/Demo	120	
		Circular base plate			
		Stiffeners			
		Multiple Stiffeners			
		Practice 9: create stiffeners and multiple stiffeners in a I section column.	**Practice**	**60**	**6**
		Practice 10: Circular base plate Base plate connections and modify its parameters	**Practice**	**60**	**6**
7	Session 7	**Macro Connection**	Theory/Demo	120	
		Shear Plate Connection			
		End Plate Connection			
		Clip Angle Connection			
		Practice 10: Create a connection with necessary macro connections	**Practice**	**120**	**6**
8	Session 8	**Macro Connection**	Theory/Demo	60	
		Splice Connection for Beam			
		Splice Connection for Column			
		Practice 11: Create a new splice connections for beam and column and modify its parameters.	**Practice**	**60**	**6**
		Macro Connection	Theory/Demo	60	
		Bolted Moment Connection			
		Welded Moment Connection			
		Practice 12: Create a new moment connections and modify its parameters.	**Practice**	**60**	**6**
9	Session 9	**Numbering**	Theory/Demo	60	
		Start Number			
		Numbering Settings			
		Perform Numbering			
		Practice 13: Perform numbering for the model.	**Practical's**	**60**	**6**
		Drawings	Theory/Demo	60	
		Types of drawings			
		General Arrangement Drawing			
		Assembly Drawing			
		Single Part Drawing			
		Multi Drawings			
		Detailing			
		Adding & Editing Dimensions			
		Adding & Editing marks			
		Adding & Editing text			
		Practice 14: Create a drawing & detail it as per industry standards.	**Practice**	**60**	**6**
10	Session 10	**Reports**	Theory/Demo	60	
		Report Generation			
		Practice 15: Generate reports as per Industry drawings	**Practice**	**60**	**6**
		Concrete Modelling	Theory/Demo	60	
		Concrete Parts: Column, beam and footing			
		Reinforcements			
		Surface treatments			
		Practice 16: Create concrete modeling as per given diagram.	**Practice**	**60**	**6**